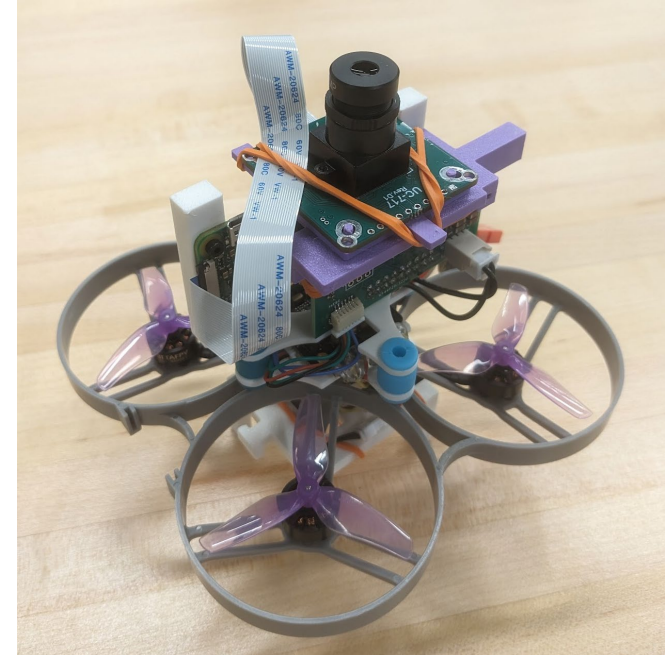
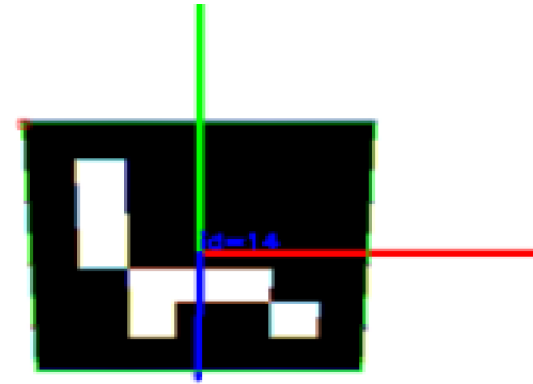


Week 8-10

Autonomy

Autonomy plan

1. Use camera to sense robot pose: x, y, z, θ , relative to fiducial tag.
 2. Send pose to user code over shared memory
 3. Generate desired roll, pitch, yaw, thrust to hold position
- Note: tags are 4x4 from <https://chev.me/arucogen/>



Building up autonomy week 8-10

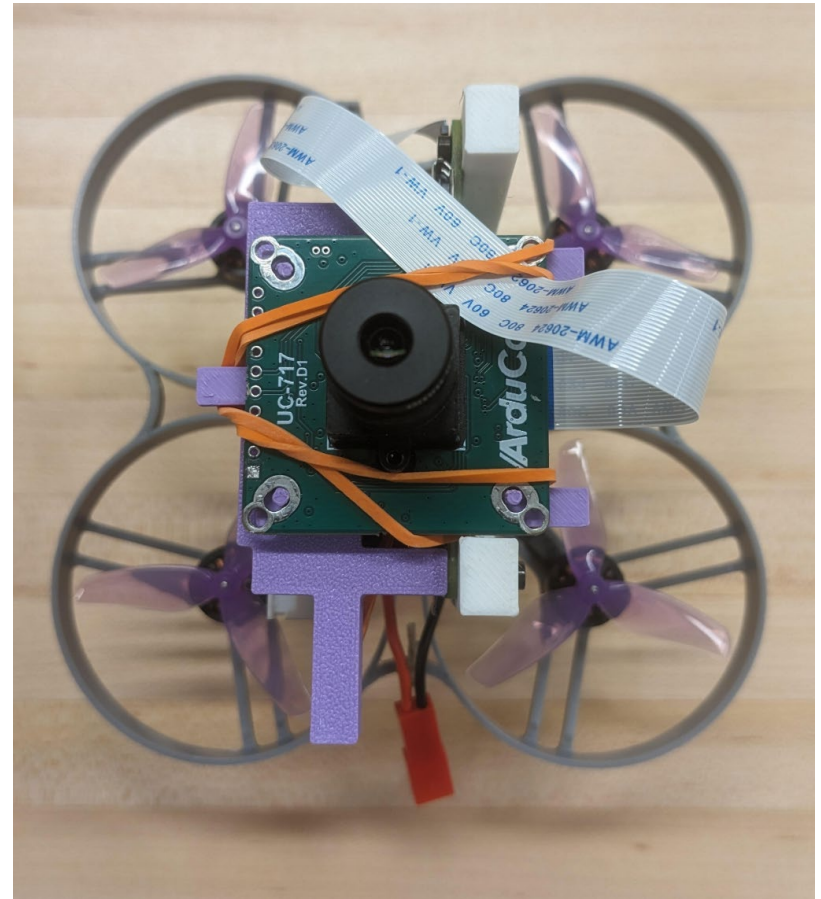
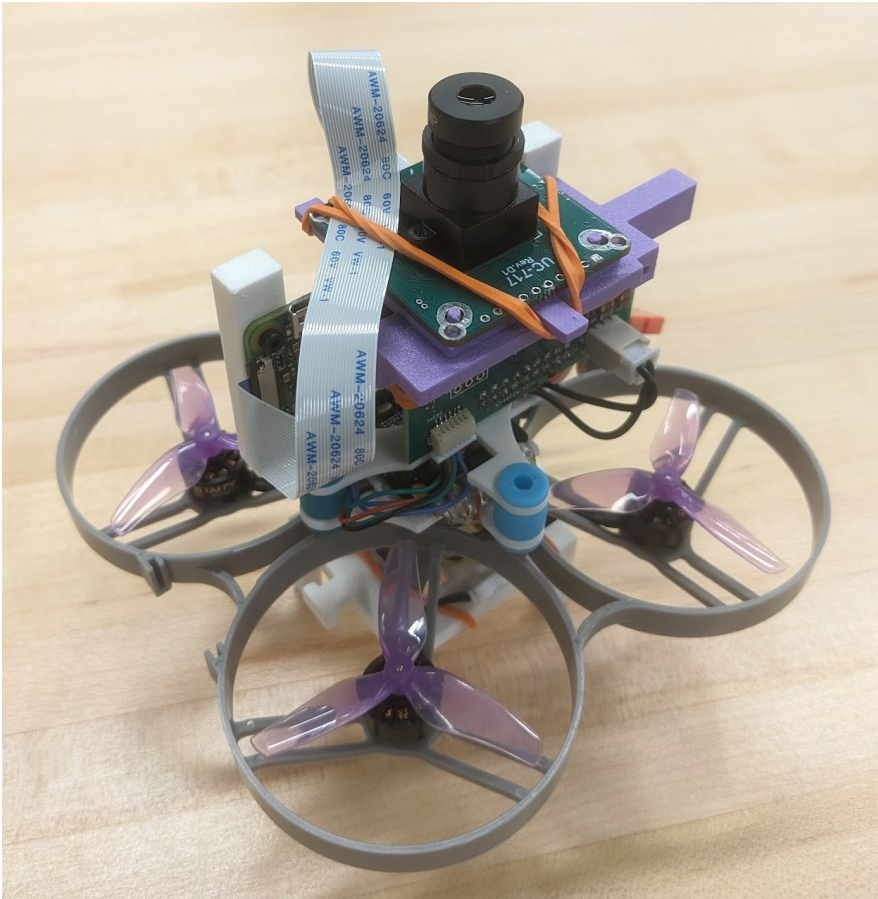
1. Camera integration
 2. Autonomous Yaw
 3. Autonomous Z (height)
 4. Autonomous X,Y
 5. FULL Autonomous flight (X,Y,Z,Yaw)
 6. FULL Autonomous flight in 6 inch sphere
- 
- Todays
class

- Milestones 1-6 are last milestones, 30 points total, All due by end of class week 10.
- Please have at least one partner pass the flight milestone before moving to autonomy

Camera integration

- Attach camera ribbon cable. CAREFULL the connector is fragile
- Ask me to check before powering up
- Use updated sd card
- Ssid: quadxxcam password: quadxxcam
- Execute camera code
 - Python3 camera.py
 - Should display x,y,z,yaw,dt,sequence number
- Look at code_updates.txt to see how to add camera shared memory to student code

Camera mounting



Milestone 1

- Show camera connected correctly (check with me)
- Power up
- Show:
 - x,y,z,yaw,sequence number displayed in student code.

Autonomous Yaw

- Goal: yaw angle from camera should be zero
- Before: the desired yaw rate was from the joystick
- Now: Desired yaw rate proportional to yaw value from camera.
- Note: now there are two P controllers for Yaw
 1. Low level: sets motor speeds based on difference between desired yaw rate and actual.
 2. High Level: Sets desired yaw rate based on difference between camera yaw and desired yaw direction.
- “X” on joystick toggles autonomy on/off

Milestone 2

- Show yaw autonomy working (in ground effect flight)
- Graph showing yaw angle from camera with a few ± 45 degree perturbations.

Z autonomy

- Goal: robot height should be kept 2 feet off ground
- Thrust is now shared autonomy:
 - $\text{Thrust} = .5 * \text{joystick_thrust} + .5 * \text{Auto_thrust}$
- Auto_thrust is computed using a PID controller based on error in Z height reported from camera

Auto thrust PID

- Note: Auto_thrust is only updated when new camera data is seen.
- $\text{Auto_thrust_P} = (\text{camera_z} - \text{z_desired}) \text{P_thrust}$
- $\text{Auto_thrust_D} =$
 $\text{D_gain_camera} * (\text{camera_z_current} - \text{camera_z_previous}) / (\text{current_time} - \text{previous_time})$
- $\text{Auto_thrust_I} += \text{I_gain_camera} * (\text{camera_z} - \text{z_desired})$
 - Try capping Auto_thrust_I to +/-100
- $\text{Auto_thrust} =$
 $\text{Auto_thrust_P} + \text{Auto_thrust_D} + \text{Auto_thrust_I}$

Milestone 3

- Autonomous control for yaw
- Shared autonomy for Z
- Human control for x,y
- Holds Z and yaw autonomously